

Lecture Notes: Integrals

Calculus I

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Contents

1	Antiderivatives	2
2	Indefinite Integrals	2
3	Basic Integration Formulas	3
4	Integration by Substitution	4
5	More Substitution Techniques	5
6	Additional Examples and Applications	6

Section 1 Antiderivatives

Definition 1 (Antiderivative) A function F is called an antiderivative of f on an interval I if $F'(x) = f(x)$ for all $x \in I$.

Theorem 1 (General Antiderivative) If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C,$$

where C is an arbitrary constant. This family of functions represents all possible antiderivatives of f .

Let G be any antiderivative of f . Then $G'(x) = f(x) = F'(x)$ for all $x \in I$. By the Mean Value Theorem corollary, $G(x) - F(x) = C$ for some constant C . Hence $G(x) = F(x) + C$.

Example 1 Find the general antiderivative of $f(x) = 3x^2$. Since $\frac{d}{dx}(x^3) = 3x^2$, an antiderivative is $F(x) = x^3$. The general antiderivative is $F(x) = x^3 + C$.

Example 2 Find the general antiderivative of $f(x) = \cos x$. Since $\frac{d}{dx}(\sin x) = \cos x$, the general antiderivative is $F(x) = \sin x + C$.

Example 3 Find the general antiderivative of $f(x) = \frac{1}{x}$ for $x > 0$. Since $\frac{d}{dx}(\ln x) = \frac{1}{x}$, the general antiderivative is $F(x) = \ln x + C$ for $x > 0$. For $x < 0$, we have $\frac{d}{dx}(\ln(-x)) = \frac{1}{x}$, so the general antiderivative on intervals not containing zero is $F(x) = \ln|x| + C$.

Remark 1 Antiderivatives are not unique. The constant C accounts for the vertical shift of the graph. Every antiderivative of f can be obtained from any other by adding a constant.

Section 2 Indefinite Integrals

Definition 2 (Indefinite Integral) The notation $\int f(x) dx$ is called the indefinite integral of f with respect to x , and it represents the most general antiderivative of f . That is,

$$\int f(x) dx = F(x) + C,$$

where $F'(x) = f(x)$ and C is an arbitrary constant. The function f is called the integrand, and C is the constant of integration.

Remark 2 The symbol $\int$ is called an integral sign, and dx indicates the variable of integration. The process of finding an indefinite integral is called integration or antidifferentiation.

Theorem 2 (Linearity of Indefinite Integrals) If f and g have antiderivatives and k is a constant, then

$$\begin{aligned} \int [f(x) + g(x)] dx &= \int f(x) dx + \int g(x) dx, \\ \int kf(x) dx &= k \int f(x) dx. \end{aligned}$$

These follow directly from the linearity of the derivative. If $F' = f$ and $G' = g$, then $(F + G)' = f + g$, so $\int (f + g) dx = F + G + C$. Similarly, $(kF)' = kf$, so $\int kf dx = kF + C$.

Example 4 Evaluate $\int (4x^3 - 2x + 5) dx$. Using linearity and the power rule,

$$\int (4x^3 - 2x + 5) dx = 4 \int x^3 dx - 2 \int x dx + 5 \int 1 dx = 4 \cdot \frac{x^4}{4} - 2 \cdot \frac{x^2}{2} + 5x + C = x^4 - x^2 + 5x + C.$$

Example 5 Evaluate $\int \left(\frac{1}{x^2} + \sqrt{x} \right) dx$ for $x > 0$. Rewrite the integrand:

$$\int (x^{-2} + x^{1/2}) dx = \frac{x^{-1}}{-1} + \frac{x^{3/2}}{3/2} + C = -\frac{1}{x} + \frac{2}{3}x^{3/2} + C.$$

Section 3 Basic Integration Formulas

Theorem 3 (Power Rule for Integration) For any real number $n \neq -1$,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C.$$

For $n = -1$,

$$\int \frac{1}{x} dx = \ln |x| + C.$$

Differentiate the right-hand side. For $n \neq -1$, $\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n$. For $n = -1$, $\frac{d}{dx} (\ln |x|) = \frac{1}{x}$ for $x \neq 0$.

Theorem 4 (Exponential and Logarithmic Formulas)

$$\begin{aligned} \int e^x dx &= e^x + C, \\ \int a^x dx &= \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1), \\ \int \ln x dx &= x \ln x - x + C. \end{aligned}$$

The first two follow from the corresponding derivative formulas. For the third, differentiate $x \ln x - x$: $1 \cdot \ln x + x \cdot \frac{1}{x} - 1 = \ln x$.

Theorem 5 (Trigonometric Integration Formulas)

$$\begin{aligned} \int \sin x dx &= -\cos x + C, \\ \int \cos x dx &= \sin x + C, \\ \int \sec^2 x dx &= \tan x + C, \\ \int \csc^2 x dx &= -\cot x + C, \\ \int \sec x \tan x dx &= \sec x + C, \\ \int \csc x \cot x dx &= -\csc x + C. \end{aligned}$$

Each formula is verified by differentiating the right-hand side. For example, $\frac{d}{dx}(-\cos x) = \sin x$, so $\int \sin x dx = -\cos x + C$.

Theorem 6 (Inverse Trigonometric Integration Formulas)

$$\int \frac{1}{1+x^2} dx = \arctan x + C,$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C.$$

These follow from the derivative formulas for arctangent and arcsine.

Example 6 Evaluate $\int (3 \sin x - 2 \sec^2 x) dx$.

$$\int (3 \sin x - 2 \sec^2 x) dx = 3(-\cos x) - 2 \tan x + C = -3 \cos x - 2 \tan x + C.$$

Example 7 Evaluate $\int \left(\frac{1}{1+x^2} + \frac{1}{\sqrt{1-x^2}} \right) dx$.

$$\int \left(\frac{1}{1+x^2} + \frac{1}{\sqrt{1-x^2}} \right) dx = \arctan x + \arcsin x + C.$$

Example 8 Evaluate $\int (2e^x + 3^x) dx$.

$$\int (2e^x + 3^x) dx = 2e^x + \frac{3^x}{\ln 3} + C.$$

Section 4 Integration by Substitution

Theorem 7 (Substitution Rule) If $u = g(x)$ is a differentiable function whose range is an interval I , and f is continuous on I , then

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

In Leibniz notation, if $u = g(x)$, then $du = g'(x) dx$, so

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

Let F be an antiderivative of f . Then $\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x)$. Thus, $\int f(g(x))g'(x) dx = F(g(x)) + C$. Also, $\int f(u) du = F(u) + C = F(g(x)) + C$. Hence the equality holds.

Remark 3 The Substitution Rule is the reverse of the Chain Rule. It is often called u -substitution. To apply it, choose u to be an inner function whose derivative also appears in the integrand (up to a constant factor).

Example 9 Evaluate $\int 2x \cos(x^2) dx$. Let $u = x^2$. Then $du = 2x dx$. The integral becomes

$$\int \cos u du = \sin u + C = \sin(x^2) + C.$$

Example 10 Evaluate $\int e^{5x} dx$. Let $u = 5x$. Then $du = 5 dx$, so $dx = \frac{du}{5}$. The integral becomes

$$\int e^u \cdot \frac{du}{5} = \frac{1}{5}e^u + C = \frac{1}{5}e^{5x} + C.$$

Example 11 Evaluate $\int \frac{1}{2x+1} dx$. Let $u = 2x+1$. Then $du = 2 dx$, so $dx = \frac{du}{2}$. The integral becomes

$$\int \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln |2x+1| + C.$$

Example 12 Evaluate $\int x\sqrt{x^2+1} dx$. Let $u = x^2+1$. Then $du = 2x dx$, so $x dx = \frac{du}{2}$. The integral becomes

$$\int \sqrt{u} \cdot \frac{du}{2} = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C = \frac{1}{3} (x^2+1)^{3/2} + C.$$

Example 13 Evaluate $\int \tan x dx$. Rewrite $\tan x = \frac{\sin x}{\cos x}$. Let $u = \cos x$. Then $du = -\sin x dx$, so $\sin x dx = -du$. The integral becomes

$$\int \frac{\sin x}{\cos x} dx = \int \frac{-du}{u} = -\ln |u| + C = -\ln |\cos x| + C = \ln |\sec x| + C.$$

Example 14 Evaluate $\int \frac{e^x}{1+e^x} dx$. Let $u = 1+e^x$. Then $du = e^x dx$. The integral becomes

$$\int \frac{du}{u} = \ln |u| + C = \ln(1+e^x) + C.$$

Section 5 More Substitution Techniques

Theorem 8 (Substitution with Definite Integrals) If g' is continuous on $[a, b]$ and f is continuous on the range of g , then

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Let F be an antiderivative of f . Then $\int_a^b f(g(x))g'(x) dx = F(g(b)) - F(g(a))$. Also, $\int_{g(a)}^{g(b)} f(u) du = F(g(b)) - F(g(a))$. The equality follows.

Example 15 Evaluate $\int_0^1 2xe^{x^2} dx$. Let $u = x^2$. Then $du = 2x dx$. When $x = 0$, $u = 0$. When $x = 1$, $u = 1$. The integral becomes

$$\int_0^1 e^u du = e^u \Big|_0^1 = e - 1.$$

Theorem 9 (Trigonometric Substitution) For integrals involving $\sqrt{a^2 - x^2}$, substitute $x = a \sin \theta$. For integrals involving $\sqrt{a^2 + x^2}$, substitute $x = a \tan \theta$. For integrals involving $\sqrt{x^2 - a^2}$, substitute $x = a \sec \theta$.

Example 16 Evaluate $\int \frac{1}{\sqrt{1-x^2}} dx$. Let $x = \sin \theta$. Then $dx = \cos \theta d\theta$. The integral becomes

$$\int \frac{\cos \theta}{\sqrt{1-\sin^2 \theta}} d\theta = \int \frac{\cos \theta}{\cos \theta} d\theta = \int 1 d\theta = \theta + C = \arcsin x + C.$$

Example 17 Evaluate $\int \frac{1}{1+x^2} dx$. Let $x = \tan \theta$. Then $dx = \sec^2 \theta d\theta$. The integral becomes

$$\int \frac{\sec^2 \theta}{1+\tan^2 \theta} d\theta = \int \frac{\sec^2 \theta}{\sec^2 \theta} d\theta = \int 1 d\theta = \theta + C = \arctan x + C.$$

Example 18 Evaluate $\int \sqrt{4-x^2} dx$. Let $x = 2 \sin \theta$. Then $dx = 2 \cos \theta d\theta$ and $\sqrt{4-x^2} = 2 \cos \theta$. The integral becomes

$$\int 2 \cos \theta \cdot 2 \cos \theta d\theta = 4 \int \cos^2 \theta d\theta.$$

Using the identity $\cos^2 \theta = \frac{1+\cos 2\theta}{2}$,

$$4 \int \frac{1+\cos 2\theta}{2} d\theta = 2 \int (1+\cos 2\theta) d\theta = 2 \left(\theta + \frac{\sin 2\theta}{2} \right) + C = 2\theta + \sin 2\theta + C.$$

Since $x = 2 \sin \theta$, $\theta = \arcsin\left(\frac{x}{2}\right)$ and $\sin 2\theta = 2 \sin \theta \cos \theta = 2 \cdot \frac{x}{2} \cdot \frac{\sqrt{4-x^2}}{2} = \frac{x\sqrt{4-x^2}}{2}$. Thus,

$$\int \sqrt{4-x^2} dx = 2 \arcsin\left(\frac{x}{2}\right) + \frac{x\sqrt{4-x^2}}{2} + C.$$

Section 6 Additional Examples and Applications

Example 19 Evaluate $\int \frac{\ln x}{x} dx$. Let $u = \ln x$. Then $du = \frac{1}{x} dx$. The integral becomes

$$\int u du = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C.$$

Example 20 Evaluate $\int \frac{1}{x \ln x} dx$ for $x > 1$. Let $u = \ln x$. Then $du = \frac{1}{x} dx$. The integral becomes

$$\int \frac{du}{u} = \ln |u| + C = \ln(\ln x) + C.$$

Example 21 Evaluate $\int \sin^3 x \cos x dx$. Let $u = \sin x$. Then $du = \cos x dx$. The integral becomes

$$\int u^3 du = \frac{u^4}{4} + C = \frac{\sin^4 x}{4} + C.$$

Example 22 Evaluate $\int \frac{x}{\sqrt{x^2+4}} dx$. Let $u = x^2+4$. Then $du = 2x dx$, so $x dx = \frac{du}{2}$. The integral becomes

$$\int \frac{1}{\sqrt{u}} \cdot \frac{du}{2} = \frac{1}{2} \int u^{-1/2} du = \frac{1}{2} \cdot 2u^{1/2} + C = \sqrt{x^2+4} + C.$$

Example 23 Evaluate $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$. Let $u = \sqrt{x}$. Then $du = \frac{1}{2\sqrt{x}} dx$, so $\frac{dx}{\sqrt{x}} = 2 du$. The integral becomes

$$\int e^u \cdot 2 du = 2e^u + C = 2e^{\sqrt{x}} + C.$$

Example 24 Evaluate $\int \frac{\cos x}{\sin^2 x} dx$. Let $u = \sin x$. Then $du = \cos x dx$. The integral becomes

$$\int \frac{du}{u^2} = -\frac{1}{u} + C = -\csc x + C.$$

Example 25 Evaluate $\int \frac{1}{e^x + 1} dx$. Rewrite the integrand by multiplying numerator and denominator by e^{-x} :

$$\int \frac{e^{-x}}{1 + e^{-x}} dx.$$

Let $u = 1 + e^{-x}$. Then $du = -e^{-x} dx$, so $e^{-x} dx = -du$. The integral becomes

$$\int \frac{-du}{u} = -\ln |u| + C = -\ln(1 + e^{-x}) + C.$$

Theorem 10 (Integration of Rational Functions by Substitution) For integrals of the form $\int \frac{f'(x)}{f(x)} dx$, the substitution $u = f(x)$ gives $\int \frac{du}{u} = \ln |f(x)| + C$.

Example 26 Evaluate $\int \frac{2x + 1}{x^2 + x + 1} dx$. Let $u = x^2 + x + 1$. Then $du = (2x + 1) dx$. The integral becomes

$$\int \frac{du}{u} = \ln |u| + C = \ln |x^2 + x + 1| + C.$$

Example 27 Evaluate $\int \frac{\sec^2 x}{\tan x} dx$. Let $u = \tan x$. Then $du = \sec^2 x dx$. The integral becomes

$$\int \frac{du}{u} = \ln |u| + C = \ln |\tan x| + C.$$

Remark 4 When using substitution, always remember to:

1. Choose u carefully so that du appears in the integrand.
2. Convert the entire integral to u and du .
3. Evaluate the integral in terms of u .
4. Substitute back to the original variable.
5. Add the constant of integration C .